

Complete MERN Stack Deployment Guide on AWS EC2

This guide explains step-by-step how to deploy a full MERN stack (MongoDB, Express, React, Node.js) application on an AWS EC2 Ubuntu server. It includes server setup, backend deployment, frontend build, and production configuration using PM2 and Nginx.

1. Launch EC2 Instance

- Go to AWS Console → EC2 → Launch Instance
- Choose Ubuntu Server 22.04
- Select instance type (t2.micro for free tier)
- Create or select a key pair (.pem)
- Allow ports: 22 (SSH), 80 (HTTP), 443 (HTTPS), 5000 (Backend)

2. Connect to EC2 via SSH

- `chmod 400 your-key.pem`
- `ssh -i your-key.pem ubuntu@YOUR_EC2_PUBLIC_IP`

3. Update Server

- `sudo apt update`
- `sudo apt upgrade -y`

4. Install Node.js

- `curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -`
- `sudo apt install nodejs -y`
- `node -v`
- `npm -v`

5. Install Git

- `sudo apt install git -y`

6. Clone Your Project

- `git clone https://github.com/yourusername/yourproject.git`
- `cd yourproject`

7. Setup Backend

- `cd backend`
- `npm install`
- Create .env file:
- `PORT=5000`
- `MONGO_URI=your_mongodb_connection`
- `JWT_SECRET=secret`

8. Install PM2 (Process Manager)

- `sudo npm install -g pm2`
- Run backend:
- `pm2 start server.js`
- `pm2 save`
- `pm2 startup`

9. Setup Frontend

- `cd ../frontend`
- `npm install`
- `npm run build`

10. Install Nginx

- `sudo apt install nginx -y`
- `sudo systemctl start nginx`

11. Configure Nginx

- `sudo nano /etc/nginx/sites-available/default`
- Example configuration:
- `server {`
- `listen 80;`
- `server_name _;`
- `location / {`
- `root /home/ubuntu/yourproject/frontend/build;`
- `index index.html;`
- `try_files $uri /index.html;`
- `}`
- `location /api {`
- `proxy_pass http://localhost:5000;`
- `}`
- `}`

12. Restart Nginx

- `sudo systemctl restart nginx`